PORTFOLIO ANALYSIS · OPEN SOURCE REUSE · KERALA

AquaSync × ICFOSS

What Kerala's open-source institute has already built — and how AquaSync should stand on it

A deep review of 54 published projects and the ~120 public repositories behind them, at icfoss.in and gitlab.com/icfoss

EVOKE 26 · Track 2, Climate Resilience & Disaster Preparedness · MACE IoT Club, Kothamangalam

54	~120	14	5
published ICFOSS projects reviewed	public repositories behind them	directly reusable by AquaSync	drop-in today, MIT-licensed

The finding, in one paragraph

ICFOSS has spent roughly a decade building the sensing and telemetry layer of a Kerala-wide environmental monitoring system, and publishing almost all of it under MIT. There are solar-powered radar water-level stations built explicitly for flood monitoring, LoRaWAN rain gauges, a river current meter, an acoustic rain gauge with a public labelled dataset, gateway designs, a network server stack, a QGIS coverage planner, a GeoNode geoportal carrying Kerala's stream network, and a live flood-alerting web portal on a canal in Thiruvananthapuram. What does not exist anywhere in that portfolio is a model that looks forward. Every ICFOSS water project answers what the level is now and whether it has crossed a line. None of them answers what the level will be in eighteen hours, or what release schedule to run to change that. AquaSync is exactly that missing layer — and it should be built to sit on top of their hardware rather than beside it.

Why this matters more than a reading list

- AquaSync's field node is currently a ₹6,250 ESP32 rig with a JSN-SR04T ultrasonic sensor. ICFOSS publishes a complete, field-deployed, solar-autonomous alternative using a VEGAPULS C 11 radar with 8 m range and a month of battery endurance — schematics, firmware and wiring diagrams included. Citing it turns a hobby build into a credible deployment plan without spending a rupee more on the demonstrator.
- The git history of this repository records that routing calibration against Neeleeswaram was blocked by data resolution. ICFOSS's LoRaWAN water current meter measures river velocity directly. That is the specific instrument the Muskingum layer needs, and it already exists.
- ICFOSS mapped 14 lakh+ 11 kV poles for KSEB and displaced a ₹200 crore proprietary estimate doing it. KSEB is the utility that operates Idukki. A working relationship with the exact counterparty AquaSync's recommendations are addressed to is worth more than any single component.
- Their Ini Njan Ozhukkate stream mapping covered 249 highland panchayats and pushed the result to OpenStreetMap. Idukki is a highland district. The channel network for this catchment is already public, and can be cross-checked against the DEM-derived geometry AquaSync computed independently.

> **The strategic reading**
>
> Treat ICFOSS not as a source of parts but as the other half of a system. They own the nervous system — sensors, radio, gateways, geodata, and a relationship with the state. AquaSync is proposing the brain. Positioning the project that way at EVOKE 26 is stronger than positioning it as a self-contained build, because it is both more honest about what is already solved and more ambitious about what is not.

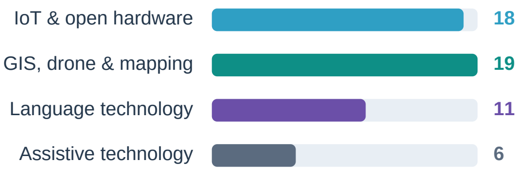
SECTION 01

What ICFOSS is, and what is actually in the portfolio

The International Centre for Free and Open Source Software is an autonomous body established by the Government of Kerala in 2009, and the state's nodal organisation for free software. It runs the Swatantra Incubator — a Kerala Startup Mission associate incubator — and hosts a LoRaWAN Centre of Excellence and an Open-FPGA Centre of Excellence, the former established with support from Hardware Mission Kerala. Practically, this means three things for a student project: the work is genuinely open-licensed, it is built against Kerala's own terrain and institutions rather than imported assumptions, and there are formal doors — incubation, the CoE, internships — that a good project can walk through.

The published portfolio

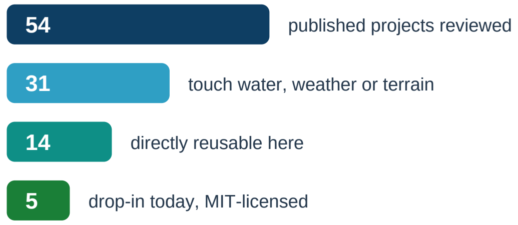
54 projects listed at icfoss.in/projects



Behind them: ~120 public repositories in 9 GitLab groups, roughly 80 of which sit in the OpenIoT group alone.

Filtered for AquaSync

How much of it actually bears on this project



The nine GitLab groups, and where the substance sits

Group	Scale	What is in it	Bearing on AquaSync
OpenIoT	~80 repos	LoRaWAN nodes, boards, gateways, energy harvesting, water level, rain, flow, air and soil sensing; ChirpStack and Grafana stacks	Decisive. Contains the field-node reference designs, the flood portal, and the network plumbing AquaSync's Layer 1 needs.
gis_and_mapping	10 repos	OpenSDI / Kerala Geoportal (GeoNode), QNetPlanner QGIS plugin, Spatial DECM browser GIS, RescueRoute, tile server, ClusterODM	High. Catchment geometry, gateway siting, evacuation routing and a lightweight map viewer for the dashboard.
drone	10 repos	Thermal imaging drone, hexacopter documentation, Kole wetland canal mapping, LiDAR configuration, RC and toolchain notes	Moderate. Post-event inundation validation and high-resolution terrain capture for the reach downstream of the dam.
Malayalam-Computing	11 repos	Malayalam OCR (incl. handwritten-text recognition), spell check, morpheme generator, root extractor, tagged corpus	Higher than it looks. Layer 4 promises Malayalam alerts — and OCR is the route to the pre-2020 bulletin data this project lacks.
Assistive_Tech	~60 repos	Braille learners, tactile Kerala map, adaptive input devices, exoskeleton, wheelchair	Low for function, useful as precedent: a rigorous open-hardware documentation pattern worth imitating in hardware/.
openfpga · incubation · Internship-projects · website	—	Chisel bootcamp and Chipyard, incubation material, internship NLP projects, the public site	Contextual. Mainly relevant as evidence of the routes by which a student project can engage the institution.

> **A caveat on sourcing**
>
> The project pages at icfoss.in are short marketing summaries; the engineering detail lives in the GitLab READMEs, and the two do not always agree. Everything asserted in this document about hardware, licensing or specifications was read from the repository itself, not from the project page. Where a repository is silent — deployment sites, calibration records, current operational status — this document says so rather than inferring.

SECTION 02

Tier 1 — the five assets AquaSync can use immediately

These are MIT-licensed, documented to the wiring-diagram level, and map one-to-one onto a component AquaSync either has as a weak placeholder or does not have at all. Each card states what the thing is, what it gives this project, and what it costs to adopt.

LAYER 1

Automatic Level Monitoring Station — radar, solar, LoRaWAN

c1_dev_lorawan_automatic_level_monitoring_station

WHAT IT IS A non-contact water level station built on the VEGAPULS C 11 radar sensor (8 m range) and ICFOSS's own C1-Dev board — an STM32 design around Murata's CMWX1ZZABZ-091 LoRaWAN module. A 100 W panel and 50 Ah Li-ion pack give over a month of endurance on battery alone, recharging from flat in about two days.

WHAT IT SENDS Water level; the last sixteen transmitted levels; rate of change in m/hr; and solar and battery voltage and current. The rate-of-change field is the notable one — it is a derived quantity computed on the node, not reconstructed later from a level series.

WHY IT MATTERS The repository states it was developed specifically as a component of a flood monitoring station. It is the same problem AquaSync's Layer 1 field node addresses, solved to a standard the ₹6,250 rig cannot reach: no contact with the water, no drift with air temperature, and no mains power.

COST TO ADOPT Zero for the demonstrator. The ESP32 rig stays as the hardware-in-the-loop model; this becomes the documented field specification, with the BOM gaining a fifth tier that is a citation rather than a purchase.

Adopt as the reference field node. It converts AquaSync's weakest claim — that a hobby ultrasonic rig represents a deployable sensor — into a sourced, costed, already-built answer.

LAYER 4

Flood Monitoring System for Technopark — the operational sibling

OpenIoT/flood-monitoring-technopark

WHAT IT IS A live flood monitoring portal for the Thettiyar canal at Technopark, Thiruvananthapuram. React, Tailwind and Chart.js on the front; Node/Express, MongoDB and JWT behind; LoRaWAN sensors via ChirpStack; deployed with Docker Compose behind Nginx. It sends email and SMS when levels cross admin-configured thresholds, and ships separate user, admin and super-admin manuals.

WHAT IT PROVES That the alerting half of AquaSync's Layer 4 — thresholds, roles, notification, historical charts — is a solved and deployed pattern in Kerala, not a speculative feature. It also shows the governance shape: a named admin sets the thresholds.

THE CONTRAST It is purely reactive. It reports the level now and alerts when a line is crossed. There is no forecast, no simulation and no release policy — because a canal has no gate to operate. This is the clearest single illustration of where AquaSync begins.

COST TO ADOPT Do not fork it. Read the three manuals and mirror their role model and threshold semantics in the Crisis Commander, and cite the system as the operational precedent AquaSync extends.

Use as precedent and as contrast: the strongest available evidence that the alerting layer is understood, and that the decision layer is the open problem.

LAYER 2

LoRaWAN Water Current Meter — the instrument routing calibration needs

ulplora_lorawan_water_current_meter…

WHAT IT IS A Savonius rotor velocity sensor on a wading rod, read by the ULP LoRa board and reported over LoRaWAN. MIT-licensed, with BOM, schematics, firmware and a setup guide.

THE PROBLEM IT SOLVES This repository's own history records an attempt to calibrate river routing against Neeleeswaram that was abandoned because the available data resolution could not resolve the wave. Muskingum's K and x are properties of a reach, and inferring them from stage alone at daily resolution is close to hopeless. Direct velocity at a known section is the measurement that makes the fit identifiable.

HONEST LIMIT One rotor at one section does not calibrate a 38 km reach. It constrains the celerity, which converts K from a free parameter into a bounded one — enough to turn an unconstrained fit into a checked one.

The single highest-value item for model credibility: it addresses a failure this project has already documented rather than a gap it has merely noticed.

LAYER 2

Acoustic Rain Gauge with edge ML — and a public labelled dataset

rainfall-acoustic-sensing · Kaggle: Rain_Da…

WHAT IT IS Rainfall estimated from sound. A USB microphone on a Raspberry Pi records timestamped WAV files; a deep model maps audio to rainfall, trained against a Davis AeroCone 6466M mechanical gauge as ground truth. Live data is published to Grafana.

THE ASSET The training data is on Kaggle as Rain_Data_Master_2023 — audio paired with mechanical-gauge rainfall, collected in Kerala. Paired rainfall observations from this state, openly licensed, are not easy to come by.

USE IN AQUASYNC The SCS-CN runoff layer is driven by forecast rainfall and validated against almost nothing local. This gives an independent local rainfall series to sanity-check depths against, and a second, cheaper sensing modality for the field node.

WATCH OUT The published accuracy of these non-mechanical instruments is modest — the companion ultrasonic anemometer reports 72.6% against its mechanical reference. Cite them as a research direction and as a data source, not as calibrated instruments.

Take the dataset now; treat the sensor as a Phase 2 research thread.

LAYER 1

Davis LoRaWAN Rain Gauge Station — deployed, and framed for dams

lorawan_based_rain_gauge_davis

WHAT IT IS A Davis AeroCone tipping-bucket gauge on the C1-Dev board, solar powered, transmitting 15-minute rainfall continuously and a daily total at 08:00. Deployed at the Greenfield Stadium site in Thiruvananthapuram. Firmware sits on ST's I-CUBE-LRWAN package; both ABP and OTAA activation are documented.

THE FRAMING The repository's own opening line describes the purpose as flood warning and reservoir management. ICFOSS built this instrument with AquaSync's use case in mind — the reservoir application is stated, the decision support to act on it is not.

USE IN AQUASYNC Pair it with the ALMS radar station to complete the field kit: rainfall in, level out, both on one LoRaWAN network, both solar, both MIT. The daily-total-at-08:00 convention also aligns neatly with the KSEB bulletin's own daily cadence.

Adopt alongside the ALMS station as the second half of the documented field specification.

Tier 1, summarised

Asset	Plugs into	Replaces or fills	Effort	Licence
ALMS radar level station	Layer 1	The ESP32 + JSN-SR04T placeholder, for field use	Cite now; build later	MIT
Davis LoRaWAN rain gauge	Layer 1	No local rainfall observation exists today	Cite now; build later	MIT
Water current meter	Layer 2	Muskingum K and x, currently unconstrained	Design work	MIT
Acoustic rain gauge dataset	Layer 2	Local rainfall ground truth for SCS-CN	Download and analyse	Kaggle terms
Flood portal (Technopark)	Layer 4	Threshold, role and notification semantics	Read the manuals	Unstated

Read the licences before you lean on them

The ALMS station, the current meter, the rain gauge and the C1-Dev board all carry MIT, which is compatible with AquaSync's own MIT licence. Several other repositories — including the Technopark flood portal — carry no licence file at all. An unlicensed public repository is not open source; it is all-rights-reserved code you can read. Cite those, learn from them, and ask before reusing.

SECTION 03

Where each ICFOSS asset attaches to AquaSync's architecture

AquaSync is built in four layers, and the ICFOSS portfolio maps onto them with a striking asymmetry. Layers 1 and 4 — getting data in, and getting decisions out to people — are densely covered by work that already exists and runs. Layer 2 is partially covered, mostly by instruments rather than models. Layer 3, the decision layer, is essentially empty. That asymmetry is the argument for this project.

LAYER 4

INTERFACE

3D twin · what-if panel · Crisis Commander · Malayalam alerts

→

WHAT ICFOSS ALREADY SUPPLIES

- Flood Monitoring Technopark — threshold alerts, SMS/email, admin roles
- Spatial DECM — browser GIS viewer beside the Three.js twin
- RescueRoute — width-aware evacuation routing
- Malayalam Computing — OCR, corpus, summarisation for alert text

LAYER 3

DECISION

Policy search over target level, start time, max release rate

→

WHAT ICFOSS ALREADY SUPPLIES

- QNetPlanner — the same multi-criteria optimisation pattern, in QGIS
- *No ICFOSS equivalent exists. This layer is AquaSync's own contribution.*

LAYER 2

SIMULATION

SCS-CN runoff · mass balance · Muskingum · tidal backwater · hydropower

→

WHAT ICFOSS ALREADY SUPPLIES

- Water Current Meter — the velocity observations Muskingum needs to calibrate
- Acoustic Rain Gauge + Kaggle dataset — labelled rainfall for the runoff model
- OpenSDI / stream mapping — catchment and channel geometry

LAYER 1

INGESTION

KSEB bulletin · IMD/Open-Meteo · INCOIS tide · Sentinel-1 · field nodes

→

WHAT ICFOSS ALREADY SUPPLIES

- ALMS radar station — VEGAPULS C 11, 8 m, solar, MIT-licensed
- Davis LoRaWAN rain gauge — 15-min and daily rainfall
- C1-Dev / ULP LoRa boards, ChirpStack, LoraLink API, gateway designs

The gap is the thesis

Across 54 published projects and roughly 120 repositories, there is no forward simulation of a water body, no optimisation over an operating policy, and no counterfactual analysis of a past event. The closest structural analogue is QNetPlanner — a QGIS plugin that selects a minimum set of gateways under weighted cost and elevation criteria. It is genuine multi-criteria optimisation, but over static infrastructure placement, not over a time-varying control policy. AquaSync's Layer 3 has no counterpart in the portfolio, which is precisely why it is worth building.

SECTION 04

Tier 2 — the hardware and network path, from expo rig to field

AquaSync's V1 rig is correct for what it is: a ₹6,250 tabletop demonstrator that closes a control loop onto a servo in a 30 cm acrylic tank. It is not a field node, and the project already says so. ICFOSS's open hardware line is what a field node looks like, and showing both — the rig you can touch, and the station you would actually install — is a stronger position than either alone.

Field node: what changes between the rig and a deployment

	AquaSync V1 rig (today)	ICFOSS field station (the target)
Controller	ESP32-WROOM-32, dual-core 240 MHz, Wi-Fi	C1-Dev v1.0 — STM32 with Murata CMWX1ZZABZ-091; or ULP LoRa
Level sensing	JSN-SR04T ultrasonic, 25–450 cm, ±1 cm, needs a DS18B20 to correct sound speed	VEGAPULS C 11 radar, 8 m, non-contact, no temperature correction needed
Link	Wi-Fi with LoRa fallback	LoRaWAN to ChirpStack, via ICFOSS gateway designs
Power	12 V 5 A mains adapter	100 W solar with MPPT and a 50 Ah Li-ion pack; >1 month of battery-only endurance
Telemetry	Level, temperature, pressure, flow	Level, last 16 levels, rate of change in m/hr, plus solar and battery voltage and current
Honest status	Built, demonstrable, closes a loop on a servo	Published and documented by ICFOSS; not built or verified by this project

The supporting network stack, all published

- LoRaWAN_in_a_Box and rpi-with-rak831-gateway — packet forwarder plus network server, and a Raspberry Pi gateway build. Enough to stand up a private LoRaWAN network without depending on anyone's infrastructure, which matters for a system that claims to work offline.
- gateway_docking_station_v1.0, solar-charge-controller-mppt and solar-battery-pack-monitoring-system-v1.0 — the unglamorous half of a remote deployment: keeping the gateway alive on a hill through a monsoon, and knowing when it will not be.
- LoraLink (api-lorawan.openiot.in) — a REST layer over ChirpStack that reads from InfluxDB rather than the LoRaWAN stack directly, with token and user administration. This is the shape of the seam between AquaSync's ingestion layer and someone else's sensor network.
- lorawan-range-mapper_v1.0 and TriLocate — coverage measurement and fine-timestamp triangulation, for verifying that a planned gateway site actually reaches the nodes you placed.

Gateway siting: a solved method for an unsolved AquaSync question

AquaSync has never answered where sensors and gateways should go in the Idukki catchment. ICFOSS has answered that class of question twice. Their viewshed methodology takes DEMs from ASF Alaska, treats candidate gateway sites as observer points at a stated antenna height, computes visibility in QGIS, and reads coverage off the result — the same approach they used to site air-quality sensors in Thiruvananthapuram and, notably, for water level monitoring at CET and Barton Hill. QNetPlanner then automates the selection: given candidate sites, a cost field normalised 1–10 and an elevation weighting, it returns a minimum set of gateways and sensors covering the area. AquaSync already derives Idukki's catchment from a DEM, so the input layer exists. This is a weekend of QGIS work that would produce a genuinely new figure: a defensible sensor network design for the catchment, with the number of gateways it needs.

A concrete, cheap, high-yield next figure

Run QNetPlanner over the existing DEM-derived Idukki catchment and publish the resulting gateway and node layout. It costs no hardware, uses a published open-source method that ICFOSS has already validated on Kerala terrain, and turns 'ESP32 field nodes' in the architecture diagram into a specific, sited, countable network.

SECTION 05

Tier 3 — geodata, routing, and an unexpected route to the 2018 data

Geospatial: the catchment is already mapped

ICFOSS asset	What it holds	Use in AquaSync
OpenSDI / Kerala Geoportal	GeoNode-based open replacement for the proprietary KSDI stack; road networks, stream networks, administrative boundaries, freely licensed	Authoritative channel and boundary geometry for the Periyar reach, and the natural place to publish AquaSync's own catchment layer
Ini Njan Ozhukkate (stream mapping)	Small watercourses across 249 highland panchayats, mapped with Nava Keralam Mission and contributed to OpenStreetMap	An independent check on the DEM-derived Idukki catchment geometry; the tributary network the runoff model implicitly assumes
Viewshed / QNetPlanner	DEM-driven visibility analysis (ASF Alaska) and a QGIS plugin for minimum-set gateway and sensor selection	Sensor network design for the catchment — see Section 04
Spatial DECM	A browser-based GIS viewer and editor; drag-and-drop GeoJSON, multiple basemaps, no installation, runs entirely client-side	A map pane for the dashboard that keeps the no-build-step, opens-from-the-filesystem constraint intact
RescueRoute	Optimal routing that accounts for vehicle size against road width, built for fire services; live tracking and reachability reporting	The evacuation half of the Crisis Commander. A release schedule implies who must move and by which road — this computes it
Thermal drone / UAV disaster work	Tarot X6 hexacopter, FLIR Vue Pro R on a 3-axis gimbal, 20 min endurance, YOLOv5 human detection, Mission Planner and QGC	Post-event validation of modelled inundation extent, and search support once a release has happened

The Malayalam stack — and the 2018 problem

AquaSync's Layer 4 promises Malayalam alerts, and ICFOSS is the single best source in the state for that: OCR (Dhriti and Lekha), a tagged corpus, a morphological analyser, a root extractor, a LibreOffice spell checker and Malayalam text summarisation, all public. Any of it improves the alert path. But there is a second, less obvious use that is worth more.

OCR as a route to the missing 2018 flood data

This project's most-stated limitation is that its dataset begins on 13 August 2020, so the 2018 flood — the reason anyone cares — cannot be modelled quantitatively. That is a limitation of one GitHub scraper, not of the historical record: KSEB and KSDMA published daily dam bulletins throughout 2018, as documents. ICFOSS's OCR stack, including a Malayalam OCR with integrated handwritten-text recognition, is built precisely to turn published Kerala documents into structured data. Recovering even a partial 2018 Idukki series would let AquaSync replay the flood everyone actually remembers — and would be a genuine contribution back, not just a borrowing. It is speculative until the bulletins are located and their quality assessed, and it should be scoped after the current deliverables are safe.

What AquaSync gives back

A reuse analysis that only takes is a weak proposal. The reciprocal case is unusually clear, because the thing AquaSync has is the thing the portfolio lacks.

- A decision layer for sensor networks ICFOSS has already deployed. Their level stations, rain gauges and flood portal produce readings and thresholds; AquaSync produces a forward simulation and a recommended action from the same inputs.
- A validated Kerala reservoir model — level–storage fitted at $r^2 = 0.9957$ over 1,836 validated rows, an October 2021 replay to 0.30 m mean error, and an independent August 2022 replay. Reusable for any Kerala reservoir with published bulletins.
- A documented data-quality finding: roughly 11% of the widely-used public Kerala dam dataset is corrupt, with storage above physical capacity between September 2020 and April 2021. Fitting on the raw feed gives $r^2 = 0.784$ against 0.996 on validated rows. Anyone else building on that feed needs to know.

- Malayalam alert templates for reservoir operations, which the Malayalam Computing group could fold into a domain corpus.

SECTION 06

What to actually do, in order

Ordered by ratio of credibility gained to time spent, and scoped so that nothing here competes with the deliverables already committed in ACTION_PLAN.md. The first three items cost no hardware and no travel.

#	Action	Why it pays	Effort	When
1	Add an ICFOSS references section to hardware/ and docs/data-sources.md	Turns the field-node claim from aspiration into a sourced, MIT-licensed, already-built specification. Pure documentation.	An afternoon	Immediately
2	Download Rain_Data_Master_2023 and cross-check local rainfall depths	The only openly licensed paired rainfall dataset from Kerala found in this review. Gives the runoff layer a local reference.	A day	This week
3	Run QNetPlanner over the DEM-derived Idukki catchment	Produces a new, defensible figure — a sited sensor and gateway network — using a method ICFOSS validated on Kerala terrain.	A weekend	Before the expo
4	Read the Technopark flood portal manuals; align Crisis Commander roles and thresholds	Matches the governance semantics of a system already running in Kerala, which is what a reviewer will ask about.	A day	Before the expo
5	Write to info@icfoss.in describing the project and the reuse	They publish an Express Interest route on every project page and run an incubator. The worst case is no reply; the best case is a conversation with the people who mapped KSEB's network.	An hour	Now — lead time is long
6	Pull the OSM stream network for the Idukki highland panchayats	Independent validation of catchment geometry this project derived from a DEM alone.	A few days	Post-expo
7	Scope the 2018 bulletin OCR recovery	Potentially removes this project's largest stated limitation. Speculative — assess document availability first.	Unknown; scope first	Post-expo
8	Explore the LoRaWAN CoE and Swatantra Incubator routes	The institutional path from an expo project to a deployed one, with Hardware Mission Kerala behind it.	Ongoing	Post-expo

Risks worth naming before you build on any of this

- Several repositories carry no licence file. Public is not open. The Technopark flood portal in particular should be cited and learned from, not forked, until its licence is clarified.
- Repository activity is uneven — some were last touched in 2024, and documentation quality varies from wiring-diagram-complete to a stub README. Verify a repository is alive before making it a dependency.
- No ICFOSS repository reviewed here documents a deployment in the Idukki catchment. Their water-level work is in Thiruvananthapuram (Thettiyar canal, CET, Barton Hill) and Kattakkada. Coverage in the Periyar basin should be treated as an open question, not an assumption.
- The non-mechanical instruments are research-grade: the ultrasonic anemometer reports 72.6% accuracy against its mechanical reference, and the acoustic rain gauge's operational accuracy is not stated. Present them as directions, not as calibrated sensors.
- Nothing here changes AquaSync's central constraint. The forecast-error study puts a real ensemble at 33-74% of the perfect-foresight cushion. Better sensors improve the present state; they do not supply the forecast.

The one-line version

ICFOSS has built Kerala's environmental sensing nervous system and given it away under MIT. Nobody has built the brain. AquaSync should be that brain, should say so plainly, and should be engineered from the outset to plug into their nerves rather than grow its own.

Sources: icfoss.in/projects and its 54 project detail pages; the icfoss GitLab organisation and its nine subgroups, read at the repository-README level; openiot.in. Reviewed 30 August 2026. Specifications quoted here come from the repositories, not the marketing pages.